BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD CHEMICAL TESTING PROCEDURE FOR BIOMEDICAL BLOOD TONIC (200ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/018**	**PAGE NO.:**	**Page 1 of 15**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Ferrous Fumarate** **Label Claim: 50mg/5ml**	**METHOD 1:** **Reagents:** a. Nitric Acid b. Hydrochloric Acid c. Dilute Sulphuric Acid d. 10% w/v solution of Ammonium thiocyanate in water. ***Standard solution:*** 40mcg/ml of Fe^{++} (as ferrous sulphate) ***Sample solution:*** To accurately weighed quantity of the sample equivalent to 50mg of Ferrous Fumarate or gluconate, add 1ml to 25ml with 1ml of Nitric acid

	DESIGNATION	SIGNATURE	DATE
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REVIEWED BY:	**QUALITY CONTROL HEAD**		
APPROVED BY:	**QUALITY ASSURANCE MANAGER**		

		and 5ml of Hydrochloric acid. Heat on a Water-bath till there are no fumes of nitrous acid. Dilute to 250ml and filter. **Procedure:** To 2ml each of the sample and standard solutions, add 5ml of Ammonium thiocyanate solution, dilute to 25ml with dilute sulphuric acid. Measure the absorption of both the Solutions at about 520nm against reagent blank. Calculte the results by comparison. % Assay of Ferrous Fumarate= $\frac{absorbance of sample}{absorbance of standard}$x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
3.	**Pyridoxine HCl** **Label Claim: 2mg/5ml**	**METHOD 1:** **Reagents:** a. 0.05% solution of p-nitroaniline: Dissolve 0.05g of p-

		nitroaniline in 5ml of methanol and dilute to 100ml with 0.033M orthophosphoric acid. b. 0.1% w/v solution of Sodium Nitrite in Water c. Phosphate Buffer, Ph 8.0: Adjust the pH of 0.5M solution of Disodium Hydrogen Phosphate to 8.0 with Sodium Hydroxide or orthophosphoric acid. d. Acetate Buffer, pH 4.25: Prepare 5% solution of sodium acetate and adust the pH to 4.25 with acetic acid. e. Mercuric Acetate solution: Dissolve 0.1589 of Mercury (iii) acetate in actetate buffer of pH 4.25 f. 0.2% solution of Sodium carbonate in Water. ***Standard solution***: 10mcg/mL of Pyridoxine Hydrochloride in water.

Sample solution:

Weigh accurately equivalent of 10mg of the substance and extract with water to produce 100ml. Further dilutions are done with water to get solution of 10mcg/ml. For oily preparations, suspend the sample in water and extract with petroleum ether (40-60°C) and use aqueous layer for analysis.

Procedure:

Take 10ml of the sample and standard solutions, add 2.5ml of phosphate buffer (8.0). In separate test tube mix 20ml of p-nitroaniline solution and 20ml of sodium nitrite solution. Add 2.5mlof this solution to 10ml each of sample and standard solutions followed by 5ml of Mercury (ii) acetate solution. Make up the volume to 25ml and measure the absorbance at 530nm against blank.

Explanatory Note

Diazotized p-nitroaniline couples with pyridoxine at position at position para to phenolic hydroxide in phosphate buffer. The complex in turn reacts with Hg++ in acetate buffer to give azo dye (λmax 530nm).

Note:

1. Since the colour of the azo dye was stable for only 10 min, the colour was stabilized by complexing with Hg++. The developed colour was stable for 24hr. Further formation of Hg++ complex resulted in bathochromic (490 to 530nm) shift and hyperchromic effect (2-fold).
2. Vitamin B1, B2, nicotinic acid, folic acid, Vitamin C and fat-soluble Vitamins did not interfere.

		% Assay of Thiamine HCl = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
4.	**Thiamine HCl** **Label Claim: 4mg/5ml**	**METHOD 1:** **Reagents:** a. 0.04%w/v solution of p-amoniphenol in ethyl alcohol. b. 0.1M Ammonium Hydroxide solution ***Standard solution:*** Dissolve 100mg of Thiamine Hydrochloride (previously dried at 100oC) in distilled water to produce 100mL. Dilute this solution suitably with water to give a concentration of 20mcg/mL ***Sample solution:*** Dissolve and dilute suitably with water to give a concentration of

		20mcg/mL **Procedure:** To 3ml each of the sample and standard solutions, add 3ml of 0.1N Ammonium Hydroxide solution and 0.5ml of p-aminophenol reagent. Wait for 5 minutes and extract the coloured complex with 10ml of Chloroform. Measure the Absorption of Organic layer at about 430nm against reagent blank. The contents of thiamine are calculated by comparison. The Method obeys Beer's law in the concentration range of 3 – 12 mcg/ml in final solution The following compounds do not interfere in the above method. The figures in brackets indicate the number of folds excess the compounds can be tolerated. • Pyridoxine

		• Niacinamide • Riboflavin • Vitamin B12 • Ascorbic Acid % Assay of Vitamin B_1 = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
5.	**Calcium D-Pantothenate** **Label Claim: 6mg/5ml**	**METHOD 1:** **Reagents:** a. 0.2M Citrate buffer, pH 5: Dissolve 21g of citric acid in 200ml of 1N sodium hydroxide and dilute to 500ml with water. Check pH 5.0 ± 0.1. b. 0.2% w/v solution of stannous chloride dihyride in citrate

buffer.

c. 2% w/v solution of ninhydin in methanol.

d. Ninhydrin-stannous chloride reagent: Mix equal volumes of reagents 2 and 3

Procedure:

After alkaline hydrolysis, the solution is made distinctly acidic with 1N sulphuric acid. Add 50ml of methanol and dilute to 100ml with water. To 2ml aliquot of the above solution add 1ml of citrate buffer and 1ml of Ninhydrin reagent. Heat in water-bath at 95-100oC for 5 min, cool and adjust to 10ml 60% Methanol. The absorbance of both sample and standard solutions is measured at about 570nm against blank and results calculated by comparison.

% Assay of Calcium D-Pantothenate = $\frac{absorbance of sample}{absorbance of standard}$ x 100

REFERENCE:

		Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
6.	**Nicotinamide** **Label Claim: 18mg/5ml**	**METHOD 1: Reagents:** a. Sulphanilic acid buffered solution: To 2.5g sulphanilic acid, add 15ml water and 2ml dilute ammonia solution. Mix, add with stirring more dilute ammonia solution if necessary to dissolve the substance, adjust pH of the solution to 4.5 with dilute ammonia b. 10% w/v solution of cyanogens bromide in water. c. Dilute ammonia solution: Dilute 2ml strong ammonia solution to 100ml with water. ***Standard solution:*** 100mcg/ml of Nicotinamide in water. ***Sample solution:*** Weigh equivalent to 10mg of the substance, suspend in water to

		make 100ml, stir and filter. **Procedure:** To 2ml each of sample and standard solutions, add 2ml of sulphanilic acid buffered solution, 5ml water, shake well and add 2ml of cyanogens bromide solution. Measure the entinction of both sample and standard solutions at 450nm after 2min against reagent blank and calculate the contents of Nicotinamide by comparison. % Assay of Nicotinamide = $\frac{absorbance of sample}{absorbance of standard}$x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
7.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method.

		- Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an “S” or “8” shape. - Allow the media to completely solidify. - Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days. **<u>TEST FOR ABSENCE OF PATHOGEN</u>**

		- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth - Transfer 10ml into 90ml of Tryptone soya broth - Incubate at 30-35^{0}c for 18-24hours - Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours - Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c - For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar. - Incubate plates at 30 to 35° for 48 hours. **A. Result Interpretation** - Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit. - If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.

		- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test. - If the colonies are of different morphology, then we can conclude that the sample contained a mixed population. Count the colonies and calculate CFU/mL by using the formula: **CFU/mL =<u>Total number of colonies obtained x dilution factor</u>** **Volume of sample used (aliquot)** - This will give the total number of viable microbial cells present in the given sample. - For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated. - If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats. - The result should be recorded following the Standard testing procedure

		SOP **BML-QDG-QA-105** - If the colonies are fused or the whole plate is covered with a single colony, then report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

*LAF--- Laminar Air Flow

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		